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Bayesian regularisation in structured additive regression: a unifying perspective on shrinkage, smoothing and predictor selection

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Abstract This paper surveys various shrinkage, smoothing and selection priors from a unifying perspective and shows how to combine them for Bayesian regularisation in the general class of structured additive regression models. As a common feature, all regularisation priors are conditionally Gaussian, given further parameters regularising model complexity. Hyperpriors for these parameters encourage shrinkage, smoothness or selection. It is shown that these regularisation (log-) priors can be interpreted as Bayesian analogues of several well-known frequentist penalty terms. Inference can be carried out with unified and computationally efficient MCMC schemes, estimating regularised regression coefficients and basis function coefficients simultaneously with complexity parameters and measuring uncertainty via corresponding marginal posteriors. For variable and function selection we discuss several variants of spike and slab priors which can also be cast into the framework of conditionally Gaussian priors. The performance of the Bayesian regularisation approaches is demonstrated in a hazard regression model and a high-dimensional geoadditive regression model.

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1 Basic concepts of Bayesian regularisation

In quite general terms, the notion of regularisation summarises approaches that allow to solve systems of equations $\boldsymbol{A\beta} \approx \boldsymbol{a}$ with respect to $\boldsymbol{\beta}$ if $\boldsymbol{A}$ is close to singular or even exactly singular. Hence, the purpose of regularisation is to introduce additional assumptions that allow to characterise useful solutions $\boldsymbol{\beta}$. In statistical terms, a typical example are linear regression models $\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathrm{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$ where $\boldsymbol{y}$ denotes a vector of responses, $\boldsymbol{\beta}$ is a q-dimensional vector of regression coefficients associated with covariates collected in the design matrix $\boldsymbol{X}$ and $\boldsymbol{\varepsilon}$ is a vector of error terms. The classical least squares estimate minimises the squared L_2 norm $\|\boldsymbol{y} - \boldsymbol{X\beta}\|_2^2$ by solving the normal equations $\boldsymbol{X'X\beta} = \boldsymbol{X'y}$ with respect to $\boldsymbol{\beta}$. Therefore, $\boldsymbol{A} = \boldsymbol{X'X}$ and $\boldsymbol{a} = \boldsymbol{X'y}$ in the general regularisation notation. If the dimension of $\boldsymbol{\beta}$ is large (possibly larger than the sample size) or if some columns in $\boldsymbol{X}$ are close to collinear, solving the normal equations becomes numerically instable. A classical regularisation approach to overcome this difficulty is to add an L_2-Tikhonov regularisation penalty to the optimisation problem, yielding

$$LS_{\mathrm{pen}}(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \to \min_{\boldsymbol{\beta}} \tag{1}$$

where $\lambda > 0$ is a regularisation parameter that determines the impact of the penalty term leading to the penalised least squares estimate $\hat{\boldsymbol{\beta}} = (\boldsymbol{X'X} + \lambda \boldsymbol{I})^{-1} \boldsymbol{X'y}$ with covariance matrix $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{X'X} + \lambda \boldsymbol{I})^{-1} \boldsymbol{X'X} (\boldsymbol{X'X} + \lambda \boldsymbol{I})^{-1}$.

In a statistical interpretation, Tikhonov regularisation corresponds to ridge estimation. For more general types

of responses and regression models we can consider regularised log-likelihood estimation by maximising penalised log-likelihoods $l_{\text{pen}}(\boldsymbol{\beta}) = l(\boldsymbol{\beta}) - \text{pen}(\boldsymbol{\beta})$, where $\text{pen}(\boldsymbol{\beta})$ corresponds to some regularisation or penalty term that measures the complexity of a solution $\boldsymbol{\beta}$.

In a Bayesian formulation of the linear model, the observation model $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ is interpreted as a conditional distribution $p(\boldsymbol{y}|\boldsymbol{\beta})$, more specifically as $\boldsymbol{y}|\boldsymbol{\beta} \sim \text{N}(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2\boldsymbol{I})$. In addition, a suitable prior distribution is assigned to $\boldsymbol{\beta}$ and (for fixed error variance σ^2) the conjugate choice is $\boldsymbol{\beta} \sim \text{N}(\boldsymbol{b}, \boldsymbol{B})$ with density $p(\boldsymbol{\beta}) \propto \exp(-\frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{b})'\boldsymbol{B}^{-1}(\boldsymbol{\beta} - \boldsymbol{b}))$. For the sake of clarity, we will treat all hyperparameters σ^2, $\boldsymbol{b}$, and $\boldsymbol{B}$ fixed for the moment, although their specification will become integral part of Bayesian regularisation in the next section. Bayes' theorem provides us with the posterior distribution

$$\boldsymbol{\beta}|\boldsymbol{y} \sim \text{N}\big((\boldsymbol{X}'\boldsymbol{X} + \sigma^2\boldsymbol{B}^{-1})^{-1}\big(\boldsymbol{X}'\boldsymbol{y} + \sigma^2\boldsymbol{B}^{-1}\boldsymbol{b}\big), \sigma^2(\boldsymbol{X}'\boldsymbol{X} + \sigma^2\boldsymbol{B}^{-1})^{-1}\big). \tag{2}$$

For the special case $\boldsymbol{b} = \boldsymbol{0}$, $\boldsymbol{B} = \tau^2\boldsymbol{I}$ (which is often chosen for simplicity in practice), the posterior simplifies to $\boldsymbol{\beta}|\boldsymbol{y} \sim \text{N}((\boldsymbol{X}'\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}'\boldsymbol{y}, \sigma^2(\boldsymbol{X}'\boldsymbol{X} + \lambda\boldsymbol{I})^{-1})$ with $\lambda = \sigma^2/\tau^2$. Hence, the posterior mean and posterior mode coincide with the penalised least squares estimate obtained from (1) while differences are observed for the covariance matrices. In general, the posterior covariance is larger than its frequentist counterpart.

In generic form, Bayesian regularisation can be formalised as follows: For a vector $\boldsymbol{y}$ of observables, the observation model is defined through a conditional distribution $p(\boldsymbol{y}|\boldsymbol{\beta}, \boldsymbol{\phi})$, where $\boldsymbol{\beta}$ is a high-dimensional parameter vector of primary interest, as for example a vector of basis function coefficients, and $\boldsymbol{\phi}$ is a low-dimensional parameter vector, comprising for example the error variance in a Gaussian regression model. Inference is regularised by specifying appropriate informative priors $p(\boldsymbol{\beta}|\boldsymbol{\vartheta})$, where the hyperparameter $\boldsymbol{\vartheta}$ includes parameters controlling complexity, such as shrinkage or smoothing parameters. The model specification is completed by assuming hyperpriors $p(\boldsymbol{\phi})$ and $p(\boldsymbol{\vartheta})$. While $p(\boldsymbol{\phi})$ is usually some simple conventional prior, $p(\boldsymbol{\vartheta})$ is typically defined hierarchically with the aim to encourage shrinkage, smoothing or selection of components of $\boldsymbol{\beta}$. Conceptually, inference is based on the posterior

$$p(\boldsymbol{\beta}, \boldsymbol{\phi}, \boldsymbol{\vartheta}|\boldsymbol{y}) \propto p(\boldsymbol{y}|\boldsymbol{\beta}, \boldsymbol{\phi})p(\boldsymbol{\beta}|\boldsymbol{\vartheta})p(\boldsymbol{\phi})p(\boldsymbol{\vartheta}).$$

The close relationship to frequentist regularisation via penalised inference is best seen if $\boldsymbol{\phi}$ and $\boldsymbol{\vartheta}$ are fixed or replaced by some plug-in values. Then, a possible estimator for $\boldsymbol{\beta}$ is again the posterior mode, i.e.

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\text{argmax}}\,\{p(\boldsymbol{y}|\boldsymbol{\beta}, \boldsymbol{\phi})p(\boldsymbol{\beta}|\boldsymbol{\vartheta})\} = \underset{\boldsymbol{\beta}}{\text{argmax}}\,\{l(\boldsymbol{\beta}) - \text{pen}(\boldsymbol{\beta})\} = \underset{\boldsymbol{\beta}}{\text{argmax}}\, l_{\text{pen}}(\boldsymbol{\beta})$$

where $l(\boldsymbol{\beta}) = \log p(\boldsymbol{y}|\boldsymbol{\beta}, \boldsymbol{\phi})$ is the log-likelihood and the penalty term $\text{pen}(\boldsymbol{\beta})$ is the negative log-prior $\log p(\boldsymbol{\beta}|\boldsymbol{\vartheta})$. Thus, the posterior mode estimate $\hat{\boldsymbol{\beta}}$ is equivalent to the penalised (log-) likelihood estimate.

However, Bayesian regularisation differs from penalised likelihood approaches by the following features: Tuning, shrinkage or smoothing parameters included in $\boldsymbol{\vartheta}$ are estimated jointly with $\boldsymbol{\beta}$. Inference for $\boldsymbol{\beta}$ is based on the marginal posterior $p(\boldsymbol{\beta}|\boldsymbol{y})$, integrating out $\boldsymbol{\vartheta}$, instead of using plug-in estimates $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\vartheta}})$ in a two-step procedure. Uncertainty in estimating regularisation or complexity parameters can be assessed through the marginal posterior $p(\boldsymbol{\vartheta}|\boldsymbol{y})$. The modular structure of hierarchical modelling and full conditionals allows to extend or to modify models and inference in a conceptually and algorithmically straightforward way. No asymptotic assumptions or conjectures are needed for finite sample inference.

This paper addresses these features for generalised semiparametric regression models with structured additive predictors of the form

$$\eta_i = \boldsymbol{x}_i'\boldsymbol{\beta} + f_1(\boldsymbol{z}_{i1}) + \cdots + f_r(\boldsymbol{z}_{ir}), \quad i = 1, \ldots, n, \tag{3}$$

where $\boldsymbol{x}$ and $\boldsymbol{\beta}$ are potentially high-dimensional vectors of covariates and parameters, while the generic functions $f_1, \ldots, f_r$ represent different types of nonlinear regression effects such as (i) a nonlinear, smooth effect of a continuous covariate x where $f_j(\boldsymbol{z}_j) = f(x)$, (ii) an interaction surface of two continuous covariates or coordinates x_1, x_2 where $f_j(\boldsymbol{z}_j) = f(x_1, x_2)$, (iii) spatial effects based on discrete spatial, i.e. regional information $s \in \{1, \ldots, S\}$ where $f_j(\boldsymbol{z}_j) = f_{\text{spat}}(s)$, (iv) varying coefficient models where $f_j(\boldsymbol{z}_j) = x_1 f(x_2)$, or (v) random effects where $f_j(\boldsymbol{z}_j) = xb_c$ with a cluster index c (see Fahrmeir et al. 2004 or Zhao et al. 2006). The predictor can be inserted in any type of regression model relying on the specification of a linear predictor by replacing this predictor with (3), see for example Fahrmeir et al. (2004), Kneib and Fahrmeir (2006) and Brezger and Lang (2006) for non-Gaussian observation models from univariate exponential families and categorical regression models or Hennerfeind et al. (2006) and Kneib and Fahrmeir (2007) for hazard regression models in the context of survival analysis. Further references will be provided in the remainder of the paper in more specific contexts.

If unknown functions are represented or approximated by basis function expansions $f_j(\boldsymbol{z}_j) = \sum_{k=1}^{K} \gamma_{jk} B_{jk}(\boldsymbol{z}_j)$ with high-dimensional basis coefficient vectors $\boldsymbol{\gamma}_j = (\gamma_{j1}, \ldots, \gamma_{jK})'$, $j = 1, \ldots, r$, the predictor (3) can be rewritten in matrix-vector notation as

$$\boldsymbol{\eta} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{Z}_1\boldsymbol{\gamma}_1 + \cdots + \boldsymbol{Z}_r\boldsymbol{\gamma}_r \tag{4}$$

where $\boldsymbol{Z}_j$ consists of the evaluations of basis functions, i.e. $\boldsymbol{Z}_j[i, k] = B_{jk}(\boldsymbol{z}_{ij})$.

Combining shrinkage and variable selection priors, previously considered for models with linear predictor only, with smoothness and selection priors for functions, we conceptualise Bayesian regularisation for a large class of regression models. As a common feature, all priors are conditionally Gaussian, given further parameters or variables regularising model complexity. Hyperpriors encourage shrinkage, smoothness and selection. This family is much larger then it may seem at first sight and comprises for example the (Bayesian) lasso, t-distributions and the elastic net for shrinkage (Sect. 2.1), and penalised splines, stationary Gaussian random fields and Gaussian Markov random fields for smoothing (Sect. 2.2). Extending the priors to finite mixture distributions leads to spike and slab selection priors that enable Bayesian variable and model selection (Sect. 3). The unifying perspective allows to link these priors to Gaussian, exponential family and hazard rate regression as a large class of structured additive regression (STAR) models and provides the possibility to construct efficient MCMC sampling schemes for joint shrinkage and smoothing. Section 4 presents applications in the context of hazard regression for survival data and a high-dimensional geoadditive regression model for predicting net rents of flats.

2 Shrinkage and smoothing based on conditionally Gaussian priors

We will now turn to the general model (4) and consider both conditionally Gaussian regularisation priors for shrinking coefficient vectors and for the estimation of smooth effects. In generic form, all priors are given by $\boldsymbol{\beta}|\boldsymbol{\vartheta}_0 \sim \mathrm{N}(\boldsymbol{b}, \boldsymbol{B})$ and $\boldsymbol{\gamma}_j|\boldsymbol{\vartheta}_j \sim \mathrm{N}(\boldsymbol{g}_j, \boldsymbol{G}_j)$ where $\boldsymbol{b} = \boldsymbol{b}(\boldsymbol{\vartheta}_0)$, $\boldsymbol{B} = \boldsymbol{B}(\boldsymbol{\vartheta}_0)$, $\boldsymbol{g}_j = \boldsymbol{g}_j(\boldsymbol{\vartheta}_j)$, $\boldsymbol{G}_j = \boldsymbol{G}_j(\boldsymbol{\vartheta}_j)$, and $\boldsymbol{\vartheta}_j$, $j = 0, \ldots, r$ are hyperparameters possibly supplemented with hyperpriors $p(\boldsymbol{\vartheta}_j)$. This specification has the particular advantage that full conditionals in a Bayesian estimation scheme resemble the form (2). It facilitates sampling based inference since a general class of samplers can be used at least for the regression coefficients (see Sect. 2.3).

2.1 Shrinkage priors

In this first part on Bayesian regularisation priors, we will focus on the parametric predictor component $\boldsymbol{\beta}$. While a noninformative prior $p(\boldsymbol{\beta}) \propto \text{const}$ is a useful standard choice (corresponding to the limiting case of a Gaussian prior with $\boldsymbol{B}^{-1} = \boldsymbol{0}$) when the dimension of $\boldsymbol{\beta}$ is low and $\boldsymbol{X}$ is well-conditioned, informative priors are considered relevant when either the dimension of $\boldsymbol{\beta}$ grows large or if $\boldsymbol{X}$ contains (close to) collinear columns. Although it may be useful to define expectations $\boldsymbol{b} \neq \boldsymbol{0}$ if prior knowledge on the expected value of a regression coefficient is available, regularisation is mostly concerned with appropriate choices of the covariance matrix $\boldsymbol{B}$ and we therefore assume $\boldsymbol{b} = \boldsymbol{0}$ for simplicity. Furthermore, for the moment we will assume $\boldsymbol{B} = \mathrm{diag}(\tau_1^2, \ldots, \tau_q^2)$, i.e. the regression coefficients are a priori independent. Although this assumption seems to be quite restrictive at first sight, the resulting class of regularisation priors is still large since quite different classes of priors are obtained depending on the prior specifications $p(\tau_j^2)$, $j = 1, \ldots, q$, for the regularisation variance τ_j^2. The reason is that different prior assumptions for τ_j^2 in the conditional Gaussian model $\beta_j|\tau_j^2 \sim \mathrm{N}(0, \tau_j^2)$ induce quite different marginal distributions

$$p(\beta_j) = \int_0^\infty p(\beta_j|\tau_j^2)p(\tau_j^2)d\tau_j^2, \tag{5}$$

leading to scale mixtures of normals. Moreover, the prior for τ_j^2 may again be formulated hierarchically, depending on further hyperparameters with suitable hyperpriors.

Ridge prior For the ridge prior, the usual smoothing parameter is given by $\lambda = 1/\tau^2$ and τ_j^2 is in principle fixed and would not be assigned a hyperprior to achieve equivalence to the classical formulation of ridge regression. However, ridge regression will usually also contain a data-driven determination of the smoothing parameter, for example based on cross validation. The Bayesian formulation allows to circumvent such external smoothing parameter selection mechanisms and instead enables joint estimation along with regression coefficients. Therefore we assign a hyperprior to τ_j^2 and an inverse gamma prior is the most common choice since it is conjugate to the Gaussian prior for β_j. Hence, we obtain the hierarchical prior formulation $\beta_j|\tau_j^2 \sim \mathrm{N}(0, \tau_j^2)$, $\tau_j^2 \sim \mathrm{IG}(a, b)$. Marginally, β_j will then follow a scaled t distribution with $2a$ degrees of freedom and scale parameter $\sqrt{a/b}$. Hence the marginal regularisation behaviour induced by a Bayesian version of the ridge prior is qualitatively quite different from the behaviour in the frequentist setting. In particular, the t-distribution has heavier tails than the Gaussian prior, leading to weaker penalisation of large coefficients.

Instead of allowing for separate smoothing parameters, we can of course also assume that $\tau_1^2 = \cdots = \tau_q^2 = \tau^2$ and that $\boldsymbol{\beta}|\tau^2 \sim \mathrm{N}(\boldsymbol{0}, \tau^2\boldsymbol{I}_q)$. Still the marginal distributions of the β_j will be given by the t distribution specified above but they will no longer be marginally independent since they implicitly depend on the same parameter τ^2.

Lasso prior The Bayesian analogue to the lasso penalty is given by the Laplace distribution. As Park and Casella (2008) point out, the Laplace distribution can be represented as a scale mixture of normals with exponential mixing distribution. To be more specific, consider the following hierarchical prior specification: $\beta_j|\tau_j^2, \lambda \sim \mathrm{N}(0, \tau_j^2)$, $\tau_j^2 \sim \mathrm{Exp}(0.5\lambda^2)$. Then, marginally β_j follows a Laplace distribution with parameter λ, i.e. $p(\beta_j) \propto \exp(-\lambda|\beta_j|)$. Still,

the smoothing parameter λ remains unspecified and in a Bayesian formulation we can simply assign a further hyperprior to it, such as a gamma distribution $\lambda^2 \sim \text{Ga}(a, b)$. In this case, the marginal distribution of β_j (marginalising over both τ_j^2 and λ) is given by

$$p(\beta_j|a,b) = \frac{a2^a}{\sqrt{\pi 2b}}\Gamma\left(a+\frac{1}{2}\right)\exp\left(\frac{\beta_j^2}{8b}\right)D_{-2a-1}\left(\sqrt{\frac{\beta_j^2}{2b}}\right),$$

where D_{-2a-1} is the parabolic cylinder function (Griffin and Brown 2005, 2007).

L_p priors Park and Casella (2008) also show that more general L_p penalties (also termed bridge regression penalties, see for example Fu 1998) with power exponential prior $p(\beta_j|\lambda) \propto \exp(-\lambda|\beta_j|^p)$ and $0 < p < 2$ can be expressed as scale mixtures of normals since

$$\exp(-|\beta_j|^p) \propto \int_0^\infty \exp\left(-\frac{\beta_j^2}{2\tau_j^2}\right)\frac{1}{\tau_j^6}s_{p/2}\left(\frac{1}{2\tau_j^2}\right)d\tau_j^2$$

where $s_p(\cdot)$ is the density of the positive stable distribution with index p. However, in this case the mixing distribution is quite complicated and it seems at least difficult to exploit the scale mixture property in Bayesian inferential procedures. Still, such a procedure would be extremely helpful, for example to approximate solutions close to L_0 regularisation, where the number of non-zero regression coefficients is penalised.

Further examples Some further examples of (frequentist) regularisation penalties can also be related to Bayesian regularisation priors. For example, the elastic net (Zou and Hastie 2005) with penalty $\text{pen}(\beta_j) = \lambda[\alpha|\beta_j| + (1-\alpha)\beta_j^2]$, $0 \le \alpha \le 1$, is simply a mixture of a Gaussian and a Laplace distribution from a Bayesian perspective. More specifically, let I_j be a binary indicator variable with $P(I_j = 1) = \alpha$. Then the conditional prior of β_j is given by

$$p(\beta_j|I_j,\lambda,\alpha) \propto I_j \exp(-\lambda|\beta_j|) + (1-I_j)\exp(-\lambda\beta_j^2)$$

where suitable hyperpriors have to be specified for the unknown hyperparameters λ and α. In a second step, the Laplace prior part of the elastic net could again be represented as a scale mixture of Gaussians so that a conditionally Gaussian prior specification results.

The elastic net has been motivated by the fact that the combination of ridge and lasso penalty tends to select groups of correlated regressors instead of a single representative. A further penalty function proposed in this context is the octogonal shrinkage and clustering algorithm for regression (OSCAR, Bondell and Reich 2008). Again, the penalty corresponds to a mixture of two components with mixture weights determined by a parameter α where the second term in the penalty corresponds to the L_∞ norm. Whether a conditionally Gaussian formulation of this penalty exists is still an open question. Other penalties such as the hard shrinkage penalty $\text{pen}(\beta_j) = \lambda^2 - (|\beta_j| - \lambda)^2\mathbb{1}(|\beta_j| < \lambda)$ or the smoothly clipped absolute deviation (SCAD, Fan and Li 2001) do not seem to have (proper) Bayesian prior counterparts since the penalties do not vanish for $|\beta_j| \to \infty$, which turns into $p(\beta_j) \nrightarrow 0$ for $|\beta_j| \to \infty$ when inverting the penalty to a prior.

General remarks Introducing hyperpriors in the Bayesian analogues to frequentist penalties in fact changes the complete interpretation of the penalty. This becomes obvious from Fig. 1 that shows the conditional log-priors $p(\beta_j|\tau_j^2)$ for Bayesian ridge and $p(\beta_j|\lambda)$ for Bayesian lasso regularisation together with the corresponding marginal log-priors. The conditional log-priors with given hyperparameters indicate the distinct difference between L_2 and L_1 penalties. In contrast, after integrating out the hyperparameters to obtain the marginal priors, the log-priors are actually quite close to each other. Both have a distinct peak at zero but the marginal lasso prior still has somewhat heavier tails than the ridge. Empirical studies (such as for example Kneib et al. 2009) provide further support that Bayesian ridge and lasso perform very similar.

Most of the penalty functionals discussed in this section have a built in absolute value penalty component and therefore may induce coefficient estimates that are exactly equal to zero in penalised likelihood or posterior mode estimation. However, Bayesian estimation in regularised regression models will often be performed with MCMC simulation algorithms. From the resulting samples, the posterior median or the posterior mean can easily be derived as point estimates but computing the posterior mode is more difficult. Since the posterior for the regression coefficients will typically be skewed, the posterior mean and median will not coincide with the posterior mode and, as a consequence, will always be different from zero. Still, the concentration around zero will typically be quite high for redundant parameters so that coefficients may well be deemed to be equal to zero from a practical perspective. Moreover, the sampling based approach provides richer information on the posterior of a regression coefficient and adequately reflects the uncertainty in estimating a parameter to be close to zero.

2.2 Smoothing priors

Now we turn to prior choices for the nonparametric effects $\boldsymbol{Z}_j\boldsymbol{\gamma}_j$, $j = 1, \ldots, r$, in STAR models. Still, we consider the class of multivariate Gaussian priors for the regression coefficients $\boldsymbol{\gamma}_j$ and most special cases will be of the form

$$p(\boldsymbol{\gamma}_j|\delta_j^2) \propto \left(\frac{1}{\delta_j^2}\right)^{\frac{\text{rank}(\boldsymbol{K}_j)}{2}} \exp\left(-\frac{1}{2\delta_j^2}\boldsymbol{\gamma}_j'\boldsymbol{K}_j\boldsymbol{\gamma}_j\right), \qquad (6)$$

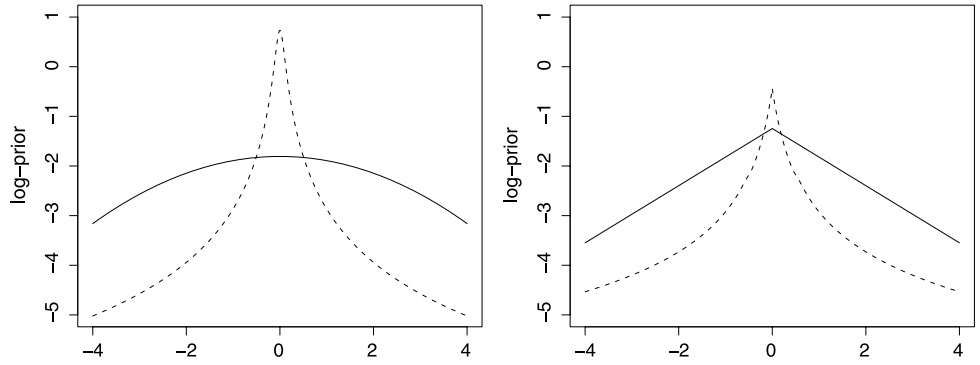


Fig. 1 Conditional (*solid lines*) and marginal (*dashed lines*) log-priors for the ridge (*left panel*) and the lasso prior (*right panel*). Hyperparameters are chosen such that roughly 90% of the probability mass are contained in the interval $[-4, 4]$, leading to $\lambda = 0.57$ for the lasso (if no prior is assigned to λ) and, e.g., $a = 0.08$, $b = 0.01$ if an additional prior for λ is used. Accordingly, the hyperparameters for the ridge are $\lambda = 0.168$ or $a = 0.28$, $b = 0.005$

i.e. $\boldsymbol{g}_j = \mathbf{0}$ and $\boldsymbol{G}_j^{-1} = \delta_j^2 \boldsymbol{K}_j$ in the notation introduced at the beginning of this section. Deriving the equivalent penalty yields $\text{pen}(\boldsymbol{\gamma}_j) = 1/(2\delta_j^2)\boldsymbol{\gamma}_j' \boldsymbol{K}_j \boldsymbol{\gamma}_j$ where the precision matrix $\boldsymbol{K}_j$ acts as a penalty matrix enforcing different types of smoothness while δ_j^2 is the smoothness parameter that determines the strength of the smoothness penalty (or the influence of the prior from a Bayesian perspective). Hyperpriors will be specified for δ_j^2 in analogy to the specification for τ_j^2 in Sect. 2.1.

Note that the prior density (6) allows for partially improper Gaussian distributions where $\boldsymbol{K}_j$ does not have full rank and can therefore not be inverted to obtain the prior covariance matrix. This partial impropriety results if specific parts of nonparametric effects shall remain unpenalised and are therefore assigned a flat, improper prior.

In the following, we will mostly drop the index j from functions, design matrices and regression coefficients to keep notation as simple as possible.

Bayesian P-splines We start our considerations with a discussion of Bayesian penalised spline smoothing for the estimation of nonparametric effects $f(x)$ with a univariate continuous covariate x following Lang and Brezger (2004), Brezger and Lang (2006) and Jullion and Lambert (2007). In this case, the function $f(x)$ is approximated in terms of K B-spline basis functions $B_k^l(x)$, $k = 1, \ldots, K$, of degree l as $f(x) = \sum_{k=1}^K \gamma_k B_k^l(x)$. Without regularisation, the resulting fit heavily depends on the number and position of the basis functions so that we are seeking suitable prior distributions for the regression coefficients γ_k to enforce the prior assumption that $f(x)$ is "smooth". From the general form of B-spline basis functions, it can be deduced that a constant curve is obtained when the coefficient sequence is close to constant. This means that prior smoothness can be achieved by assuming that $\gamma_k \approx \gamma_{k-1}$, which can be expressed more formally by assigning a prior to the difference of adjacent coefficients, i.e. $\gamma_k - \gamma_{k-1}|\delta^2 \sim \text{N}(0, \delta^2)$. Formally this is equivalent to assuming a first order random walk $\gamma_k = \gamma_{k-1} + u_k$, $k = 2, \ldots, K$ with $u_k \sim \text{N}(0, \delta^2)$ and $p(\gamma_1) \propto \text{const}$. The prior specification reflects that we want to penalise deviations from a constant but that the exact value of this constant is arbitrary and therefore supplemented with a flat, noninformative prior. Of course, higher order differences can be considered in complete analogy, leading for example to $\gamma_k - 2\gamma_{k-1} + \gamma_{k-2}|\delta^2 \sim \text{N}(0, \delta^2)$ and $p(\gamma_1, \gamma_2) \propto \text{const}$ which is equivalent to a second order random walk. From the properties of B-splines it follows that a random walk of order d with flat priors for initial values leaves a polynomial of order $d - 1$ in x unpenalised. These random walk priors are the Bayesian version of difference penalties in the P-splines approach of Eilers and Marx (1996).

From the conditional distributions induced by the random walk prior and the noninformative priors for the initial values, we can deduce the joint prior distribution for the vector of regression coefficients $\boldsymbol{\gamma}$. This joint prior is of the form (6) with precision or penalty matrix $\boldsymbol{K} = \boldsymbol{D}'\boldsymbol{D}$, where $\boldsymbol{D}$ is a d-th order difference matrix. The precision is rank deficient with $\text{rank}(\boldsymbol{K}) = K - d$ for a d-th order random walk prior. A further choice that has to be made is a hyperprior for the smoothing variance δ^2 and the inverse gamma distribution $\delta^2 \sim \text{IG}(a, b)$ is conjugate to the Gaussian prior for $\boldsymbol{\gamma}$ resulting in simple Gibbs updates for δ^2.

Alternative P-splines and extensions Penalised splines have also sometimes been introduced based on other bases than B-splines. The most prominent example are P-splines based on the truncated power series basis (TP basis) of degree l, where

$$f(x) = \sum_{k=1}^{l} \gamma_k x^k + \sum_{k=l+1}^{K} \gamma_k (x - \kappa_k)_+^l \tag{7}$$

and $\kappa_1, \ldots, \kappa_K$ are a sequence of knots partitioning the covariate domain. In a Bayesian formulation, priors for coefficients are $p(\boldsymbol{\gamma}_1) \propto \text{const}$ and $p(\boldsymbol{\gamma}_2) \sim \text{N}(\mathbf{0}, \delta^2 \boldsymbol{I})$, i.e. a flat prior for the polynomial coefficients $\boldsymbol{\gamma}_1 = (\gamma_1, \ldots, \gamma_l)'$ and a Gaussian ridge prior for the truncated polynomial coefficients $\boldsymbol{\gamma}_2 = (\gamma_{l+1}, \ldots, \gamma_K)'$. This prior specification also suggests an interpretation of smoothing approaches as mixed models (see Wand 2003 and Ruppert et al. 2003 for details). In fact, such a mixed model representation can be derived quite generally for any type of regression effect with a Gaussian prior of the form (6), see Fahrmeir et al. (2004), which has several advantages. First of all, mixed model methodology can be employed for determining the smoothing variance δ^2 since it simply turns into a variance component in the mixed model. Moreover, it is also of theoretical value and allows, for example, to study the propriety of posteriors in structured additive regression models (Fahrmeir and Kneib 2009) or to derive linear constraints on $\boldsymbol{\gamma}$ that turn the partially improper prior into a proper one (Panagiotelis and Smith 2008). We will also use the mixed model representation of P-splines in the following paragraph to derive connections between nonparametric smoothing and correlated responses.

Adaptive and robust penalised spline variants can be derived by modifying the random walk and in particular the hyperprior specification for the smoothing variance δ^2. Most proposals in this direction either assume that δ^2 is replaced by a sequence of independent prior variances δ_k^2, $k = 2, \ldots, K$, or that the variance varies itself as a function of x, i.e. $\delta^2 = \delta^2(x)$ which is again represented in basis functions. An example for the first approach is given in Lang and Brezger (2004) who assume that δ_k^2 is i.i.d. $\text{Ga}(\frac{\nu}{2}, \frac{\nu}{2})$. The second approach is, for example, pursued in Baladandayuthapani et al. (2005). A compromise that is based on a piecewise constant variance function with automatic determination of the jump points is available in Scheipl and Kneib (2009). All these variants allow to keep the conditionally Gaussian distribution for the spline coefficients $\boldsymbol{\gamma}$, where only the precision matrix has to be altered.

Smoothing and correlation We will now use the mixed model interpretation of penalised splines to discuss identifiability constraints in semiparametric regression models including smooth time trends and temporally correlated error terms. Consider time series data following the model

$$y_t = f(t) + \varepsilon_t, \quad t = 1, \ldots, T, \tag{8}$$

where $f(t)$ is represented as a penalised spline, leading to $\boldsymbol{y} = \boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$. In mixed model representation, an equivalent specification is given by

$$\boldsymbol{y} = \boldsymbol{Z}_1\boldsymbol{\gamma}_1 + \boldsymbol{Z}_2\boldsymbol{\gamma}_2 + \boldsymbol{\varepsilon}, \qquad p(\boldsymbol{\gamma}_1) \propto \text{const},$$
$$\boldsymbol{\gamma}_2 \sim \text{N}(\mathbf{0}, \delta^2 \boldsymbol{I})$$

and, from the mixed model perspective, there are two interpretations of such a model: Conditional on the "random" effect $\boldsymbol{\gamma}_2$, the induced distribution for the responses is $\boldsymbol{y} \sim \text{N}(\boldsymbol{Z}_1\boldsymbol{\gamma}_1 + \boldsymbol{Z}_2\boldsymbol{\gamma}_2, \sigma^2 \boldsymbol{I})$. Marginally, we obtain $\boldsymbol{y} \sim \text{N}(\boldsymbol{Z}_1\boldsymbol{\gamma}_1, \delta^2 \boldsymbol{Z}_2\boldsymbol{Z}_2' + \sigma^2 \boldsymbol{I})$, i.e. a general linear model with non-diagonal covariance matrix. Hence, the temporal trend induces correlations in the response variables. Moreover, we can derive the exact form of the induced correlation function from the covariance matrix $\sigma^2 \boldsymbol{I} + \delta^2 \boldsymbol{Z}_2\boldsymbol{Z}_2'$ by rowwise standardisation. Figure 2 illustrates this for simulated data with $f(t) = \sin(2\pi(t-1)/T)$, $t = 1, \ldots, T = 601$ and both B-spline based and truncated power series based P-splines. While B-splines lead to an almost symmetric correlation function, the correlation function for the truncated power series is extremely asymmetric with long ranged correlation to the right. This corresponds to a specific feature of the truncated power series basis where a basis function is zero up to a certain knot and remains positive from this point on. In contrast, B-splines are defined locally, leading also to a more localised correlation structure. The spikes in the correlation functions result from the additional inclusion of the error variance.

Since the specification of a P-spline for the temporal effect already allows to deal with correlations in the data, identifiability problems may arise when additional temporally correlated errors $\boldsymbol{\varepsilon} \sim \text{N}(\mathbf{0}, \boldsymbol{\Sigma})$ are included in a temporal or, more generally, a longitudinal model. From a Bayesian perspective, both the error and the nonparametric function are assigned stochastic process priors that are only differentiated by prior assumptions about their smoothness, see also Fahrmeir and Kneib (2008).

Spatial smoothness priors Several types of spatial smoothness priors can also be cast in the conditional Gaussian prior framework (6). In case of regional data, (intrinsic) Gaussian Markov random fields extend the random walk prior to two dimensions, see for example Besag et al. (1991) and in particular Rue and Held (2005). In case of point-referenced data, bivariate extensions of P-splines (Lang and Brezger 2004) and stationary Gaussian random fields (Kammann and Wand 2003; Fahrmeir et al. 2004) are possible choices. The latter can also be interpreted as a basis functions approach with radial basis functions, further supporting the relation between (spatial) smoothing and the modelling of (spatial) correlations (Nychka 2000).

2.3 Bayesian inference

In principle, different viable strategies for Bayesian inference exist for the class of STAR models with conditionally Gaussian priors. Simulation based inference utilising MCMC techniques allows to fully exploit the hierarchical

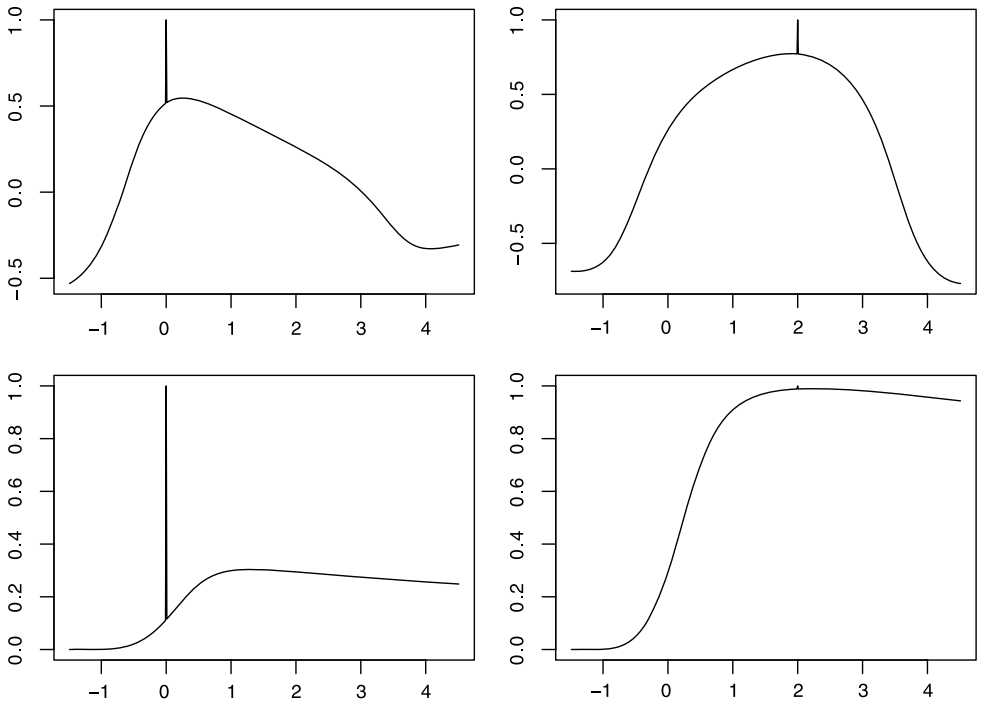


Fig. 2 Correlation function induced by the marginal perspective on the nonparametric model (8) for P-splines based on a B-spline basis (*top row*) and a truncated power series basis (*bottom row*). The *left* and the *right panel* correspond to a value early in the time series and in the middle of the time series

structure induced by the conditional prior specification. Still, approximate solutions may be valuable alternatives that circumvent the difficulties of monitoring mixing and convergence of Markov chains. Such solutions can for example be based on the mixed model representation of nonparametric smoothers in combination with the Laplace approximation for obtaining marginal posterior estimates of the smoothness variance (see for example Fahrmeir et al. 2004). Improved approximation accuracy can be achieved with integrated nested Laplace approximations as recently proposed in Rue et al. (2009). However, both approximate solutions have the drawback that they are difficult to extend to situations where hierarchical stages are added to the hyperparameters of the prior for the regularisation variances. Such situations are, for example, relevant for the Bayesian lasso or in case of adaptive smoothing.

Since all priors for regression coefficients are conditionally Gaussian, general updating schemes can be developed regardless of the specific type of effect represented by these coefficients. We will illustrate these general schemes for the parametric effects with $\boldsymbol{\beta}|\boldsymbol{\vartheta}_0 \sim \mathrm{N}(\boldsymbol{b}, \boldsymbol{B})$. For Gaussian responses or responses that can be related to latent Gaussians (such as the probit model), the full conditional for $\boldsymbol{\beta}$ is a Gaussian distribution $\boldsymbol{\beta}|\cdot \sim \mathrm{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta)$ where

$$\boldsymbol{\mu}_\beta = \boldsymbol{\Sigma}_\beta \left(\frac{1}{\sigma^2} \boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{\eta}_{-\beta}) + \boldsymbol{B}^{-1}\boldsymbol{b} \right),$$

$$\boldsymbol{\Sigma}_\beta = \left(\frac{1}{\sigma^2} \boldsymbol{X}'\boldsymbol{X} + \boldsymbol{B}^{-1} \right)^{-1},$$

$\boldsymbol{\eta}_{-\beta} = \boldsymbol{\eta} - \boldsymbol{X}\boldsymbol{\beta}$ and σ^2 equals the error variance. In case of non-Gaussian responses, the full conditional does no longer have a closed form and therefore requires Metropolis Hastings (MH) updates. Approximating the full conditional with a Gaussian distribution yields a proposal density that automatically adapts to the location and curvature and therefore usually yields excellent mixing properties (see Brezger and Lang 2006 for details). Exactly the same type of updates and proposals can be utilised for nonparametric functions. In this case, however, a special structure arises for the full conditional covariance matrix (or the covariance matrix of the proposal density) and sparse matrix computations can be used to speed up the run time of MCMC algorithms considerably (see Brezger and Lang 2006 for details).

The advantage of MCMC samplers for hierarchical prior specifications becomes obvious when turning to the hyperparameters. Since these are (conditionally) separated from the data by the regression coefficients, exactly the same type of full conditional arises regardless of the response distribution. For example, in case of the Bayesian ridge, we obtain $\tau_j^2|\cdot \sim \mathrm{IG}(a + 0.5, b + 0.5\beta_j^2)$ for separate smoothing variances and $\tau^2|\cdot \sim \mathrm{IG}(a + 0.5q, b + 0.5\sum_{j=1}^q \beta_j^2)$ for one single variance parameter. For the Bayesian lasso, the full conditionals are given by

$$\frac{1}{\tau_j^2}\bigg|\cdot \sim \mathrm{InvGauss}\left(\frac{|\lambda|}{|\beta_j|}, \lambda^2 \right),$$

$$\lambda^2|\cdot \sim \mathrm{Ga}\left(a + q, b + \frac{1}{2}\sum_{j=1}^q \tau_j^2 \right).$$

In both cases, closed form full conditionals arise and Gibbs sampling steps can be employed. In particular, the scale mixture representation of the Bayesian lasso turns out to be useful to replace the numerically cumbersome Laplace prior into a conditionally Gaussian prior with Gibbs updates for the hyperparameters. For smoothing variances, we also obtain Gibbs updates from the full conditional inverse gamma distribution $\delta_j^2|\cdot \sim \text{IG}(a_j + 0.5\,\text{rank}(\boldsymbol{K}_j), b_j + 0.5\boldsymbol{\gamma}_j'\boldsymbol{K}_j\boldsymbol{\gamma}_j)$.

3 Predictor selection using spike and slab priors

Predictor selection in structured additive regression models comprises the following tasks: Variable selection, i.e. inclusion or exclusion of covariates x_j with linear effects $\beta_j x_j$ as part of the linear predictor $\boldsymbol{x}'\boldsymbol{\beta}$, selection of functions $f_j(x_j)$ and, for clustered or longitudinal data, selection of a parsimonious (Gaussian) random effects prior through a parsimonious choice of the covariance matrix of i.i.d. random effects. This section describes approaches and concepts based on spike and slab priors, allowing for jointly estimating and selecting the various components in a structured additive predictor. Spike and slab priors are discrete mixture priors, formalizing the idea that a regression coefficient or a variance component is significant or (practically) non-significant. We describe the three selection issues in turn, but they can be combined for jointly selecting and estimating components of interest.

3.1 Variable selection

Bayesian variable selection is commonly based on spike and slab priors for regression coefficients, and it has been most extensively discussed for the classical linear Gaussian model and for the closely related case of Gaussian nonparametric regression where a function is approximated through a linear combination of basis functions. We focus on approaches where selection priors can be formulated in form of conditionally Gaussian densities, fitting into the same general framework as shrinkage and smoothness priors in Sect. 2. This allows to extend existing approaches to STAR models for Gaussian and non-Gaussian responses and to incorporate selection modules into existing MCMC algorithms for shrinkage and smoothing.

The basic idea for a Gaussian linear model with $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_q)$, where β_0 is the intercept, is as follows (George and McCulloch 1993, 1995, 1997): Each component β_j of $\boldsymbol{\beta}$ is modelled as having come either from a distribution with most (or all) of its mass concentrated around zero (the "spike"), or from a comparably diffuse distribution with mass spread out over a large range of values (the "slab"). Introducing binary latent indicators $J_j \in \{0, 1\}$, this prior can be formally expressed as a mixture

$$p(\beta_j|J_j) = (1 - J_j)\varphi(\beta_j; 0, v_{0j}^2) + J_j\varphi(\beta_j; 0, v_{1j}^2)$$

of Gaussian densities with variances $v_{0j} \ll v_{1j}$, and $\mathbb{P}(J_j = 1|w_j) = 1 - \mathbb{P}(J_j = 0|w_j) = w_j$. The idea is that if $J_j = 0$ and v_{0j} is close to zero, then β_j is likely to be close to zero and the effect of the covariate x_j is "practically" non-significant. If $J_j = 1$ and v_{1j} is large, then β_j is significant. George and McCulloch (1993, 1997) give guidelines for choosing the hyperparameters v_{0j}^2 and v_{1j}^2, including a semi-automatic approach. The hyperparameter $w_j = \mathbb{P}(J_j = 1|w_j)$ is the prior probability that x_j has a significant effect and should be included in the model. Usually, the intercept term β_0 is always included so that $w_0 = 1$. Excluding this degenerate case, the standard joint prior for $\boldsymbol{J} = (J_1, \ldots, J_q)$ is the independent Bernoulli prior

$$p(\boldsymbol{J}|\boldsymbol{w}) = \prod_{j=1}^{q} w_j^{J_j}(1 - w_j)^{1-J_j}. \tag{9}$$

A particularly simple choice is to assume $w_j = w = 0.5$ for all $j = 1, \ldots, q$, resulting in the seemingly uninformative, discrete uniform prior $p(\boldsymbol{J}) = 2^{-q}$. However, the number $q_{\boldsymbol{J}} = \sum_j J_j$ of significant effects has a binomial distribution $\text{B}(q, 0.5)$, so that $\text{E}(q_{\boldsymbol{J}}|w = 0.5) = q/2$. This may be unreasonable for high-dimensional $\boldsymbol{\beta}$. A more flexible approach is to place a hyperprior on w. A beta prior $p(w) \propto w^{a_w-1}(1 - w)^{b_w-1}$ is a reasonable and convenient choice, including the uniform prior $w \sim \text{U}(0, 1)$ as a special case for $a_w = b_w = 1$. Generally, a_w and b_w can be chosen so that they match desired values $\text{E}(q_{\boldsymbol{J}})$ and $\text{Var}(q_{\boldsymbol{J}})$, see Kohn et al. (2001).

For joint selection of components of $\boldsymbol{\beta}$ and estimation of functional or random effects in structured additive regression models, we summarise as follows: The diffuse prior $\boldsymbol{\beta} \sim N(\boldsymbol{0}, \boldsymbol{B})$ is replaced by the hierarchy $\boldsymbol{\beta}|\boldsymbol{J} \sim \text{N}(\boldsymbol{0}, \boldsymbol{D}_{\boldsymbol{J}})$, where $\boldsymbol{D}_{\boldsymbol{J}} = \text{diag}((1 - J_j)v_{0j}^2 + J_j v_{1j}^2)$. In addition, we set $J_0 \equiv 1$ for always including the intercept term β_0 and choose the Bernoulli prior (9) with a beta prior for w.

Extending Gibbs sampling for Gaussian models and MCMC algorithms with IWLS proposals for non-Gaussian models to include variable selection is conceptually straightforward. To draw from full conditionals $\boldsymbol{\beta}|\cdot$ for regression coefficients (given $\boldsymbol{J}$, all other parameters and $\boldsymbol{y}$) the matrix $\boldsymbol{B}$ appearing in Gaussian full conditionals or IWLS proposals is replaced by $\boldsymbol{D}_{\boldsymbol{J}}$. The indicator variables J_j are drawn componentwise from Bernoulli distributions with probability $\mathbb{P}(J_j = 1|\beta_j) = A_j/(A_j + B_j)$, where $A_j = \varphi(\beta_j; 1, v_{1j}^2)w$ and $B_j = \varphi(\beta_j; 0, v_{0j}^2)(1 - w)$. The weight w is drawn from $w|\cdot \sim \text{Be}(\tilde{a}_w, \tilde{b}_w)$ with updated hyperparameters $\tilde{a}_w = a_w + q_{\boldsymbol{J}}$, $\tilde{b}_w = b_w + (q - q_{\boldsymbol{J}})$. As a key

advantage of MCMC inference, all other full conditionals remain untouched.

To avoid the choice of a suitably small hyperparameter v_{0j}^2 at all, it appears natural to consider the limiting case $v_{0j}^2 = 0$. This would lead to the prior

$$\beta_j | J_j \sim (1 - J_j)\delta_0(\cdot) + J_j \, \mathrm{N}(0, v_{1j}^2),$$

where $\delta_0(\cdot)$ denotes a point mass at 0. In this formulation, $J_j = 0$ indicates that $\beta_j = 0$, that is, x_j is excluded from the linear predictor, and, vice versa, $J_j = 1$ indicates $\beta_j \neq 0$. However, then the Gibbs or MH sampling schemes as outlined above cannot be used anymore because they generate reducible, nonconvergent Markov chains. Without any modification, the sampler gets stuck when a value $\beta_j = 0$ is drawn. Geweke (1996) proposes an alternative that seems to offer improved convergence rates at the expense of computational speed.

For *Gaussian models*, a different strategy based on Zellner's g-prior, i.e. a Gaussian prior with covariance matrix $g\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$, $g > 0$, leads to computationally very efficient sampling schemes, see e.g. Smith and Kohn (1996) and George and McCulloch (1997). A key feature is that this conjugate prior setting yields rapidly computable closed form expressions for $p(\boldsymbol{y}|\boldsymbol{J})$ by analytically integrating out $\boldsymbol{\beta}$ and σ^2. This allows for efficient updating schemes and for direct computation of Bayes factors and posterior model probabilities $p(\boldsymbol{J}|\boldsymbol{y})$. Because the hyperparameter g acts as a complexity parameter, its choice can be critical. Various choices and hyperpriors for g have been suggested, see for example Fernandez et al. (2001), Ley and Steel (2009) and Liang et al. (2008) for a review and suggestions for mixtures of g-priors. Avoiding the g-prior approach, Casella and Moreno (2006) propose intrinsic priors for objective Bayesian variable selection. Extensions of the g-prior approach to non-Gaussian regression are not straightforward because the marginal $p(\boldsymbol{y}|\boldsymbol{J})$ can no longer be obtained automatically. Wang and George (2007) propose an integrated Laplace approximation for generalised linear models with a GLM version of the g-prior. Conceptually, the g-prior setting can also be extended to non-Gaussian models that are formulated in terms of latent auxiliary Gaussian models in a further stage of the hierarchy, such as for binary or multicategorical probit models. For other non-Gaussian models, such as count data regression, analogous extensions could be based on auxiliary models with error distributions approximated through a finite mixture of normals (Frühwirth-Schnatter and Wagner 2006; Frühwirth-Schnatter et al. 2008).

Instead of encouraging variable selection by placing priors directly on the regression coefficients, a different strategy is to place spike and slab priors on the variances τ_j^2 in the conditional priors $p(\beta_j|\tau_j^2)$ for regression coefficients. This concept is formalised through the shrinkage prior of Ishwaran and Rao (2003, 2005), introduced for linear models again. Their hierarchical prior for $\beta_j|\tau_j^2$ is

$$\beta_j|\tau_j^2 \overset{\text{ind.}}{\sim} \mathrm{N}(0, \tau_j^2), \quad j = 1, \ldots, q,$$

$$\tau_j^2|J_j \sim (1 - J_j)\,\mathrm{IG}(a_\tau, \nu_0 b_\tau) + J_j\,\mathrm{IG}(a_\tau, \nu_1 b_\tau)$$

with hyperparameters $0 < \nu_0 \ll \nu_1$, ν_0 close to zero. The binary indicators $J_j \in \{0, 1\}$ are related to the indicators $I_j \in \{\nu_0, \nu_1\}$ in the original formulation through $J_j = 1 \Leftrightarrow I_j = \nu_1$, $J_j = 0 \Leftrightarrow I_j = \nu_0$. Together with the prior (9) for the indicators, we have $\tau_j^2|w \sim (1 - w)\,\mathrm{IG}(a_\tau, \nu_0 b_\tau) + w\,\mathrm{IG}(a_\tau, \nu_1 b_\tau)$, i.e. the prior for τ_j^2 is a spike and slab prior for variances specified as a bimodal mixture of two inverse Gamma distributions. Therefore, we call this structure normal mixture of inverse gamma (NMIG) prior. Integrating out τ_j^2, the marginal distribution of β_j is a mixture of two scaled t-distributions.

Thus, the inverse Gamma spike and slab prior for variances induces a spike and slab prior for regression coefficients. The issue of specifying the variances v_{0j}^2 and v_{1j}^2 in the bimodal Gaussian mixture prior is now transferred to the choice of ν_0 and ν_1. For Gaussian models, Ishwaran and Rao (2005) suggest default options, in particular $\nu_1 = 1$, after standardizing all covariates and rescaling the model, see also Konrath et al. (2008) for extensions to non-Gaussian and hazard rate models.

This NMIG variance prior approach provides a natural procedure for selecting covariates, using posterior samples of the indicator variables: The larger the estimated posterior probability of an indicator variable (i.e. the percentage of ν_1 values in the sample), the larger is the evidence that the corresponding covariate has non-negligible effects, and can be eliminated from the model.

3.2 Function selection

The following related issues are of interest in function selection: First, should a functional effect $f_j(x_j)$ be included in the predictor at all? Because we always include a common intercept β_0, we assume that all functions are centered around zero. Therefore, we want to know whether $f_j(x_j)$ differs significantly from 0. Second, we are often more interested in the following question: Is the effect of a covariate x_j linear, that is $f_j(x_j) = \beta_j x_j$, or is it nonlinear, i.e. differs $f_j(x_j)$ significantly from a straight line? In this case, we may decompose the function into a linear and a nonlinear part, i.e. $f_j(x_j) = \beta_j x_j + \tilde{f}_j(x_j)$, where the nonlinear part solely captures deviations from a linear effect. More generally, we may be interested whether a functional effect differs from a polynomial of higher degree. For identifiability, functions $\tilde{f}_j(x_j)$ have to be "orthogonal" to the space of linear

functions or, more generally, of polynomials. How to best approach these issues is also related to the choice of basis functions and smoothness priors for basis function coefficients.

This is most immediately seen when a function is represented by a linear combination of truncated power basis functions as in (7). For a linear spline ($l = 1$) with knots κ_k, the effect of x_j is exactly linear for $\boldsymbol{\gamma}_j = (\gamma_{2j}, \ldots, \gamma_{Kj})' = \mathbf{0}$, and it is "practically" linear if $\boldsymbol{\gamma}$ is "practically" zero. This leads immediately to the idea to transfer concepts from variable selection to function selection by introducing corresponding spike and slab priors directly for the basis function coefficients or, alternatively, for the variance parameters δ_j^2 in the conditional Gaussian prior (6) where $\boldsymbol{K}_j = \boldsymbol{I}$ for the TP basis representation (7). If we want to explore if x_j has a significant effect $f_j(x_j) \neq 0$ at all, we have to combine this with a selection prior for γ_{1j}.

If, instead, we represent $f_j(x_j)$ through a B-spline basis with a partially improper penalty matrix $\boldsymbol{K}_j$, choice of selection priors is less immediate: Improper priors should be avoided for variable selection problems, and function selection is also related to the null space of $\boldsymbol{K}_j$, i.e. to functions that are not penalised through $\boldsymbol{K}_j$. Several strategies are possible to circumvent problems arising with (partially) improper priors for Bayesian P-splines: First, additional restrictions or proper priors can be imposed on initial values of parameters as in Chib and Jeliazkov (2006) or, more symmetrically, on initial and final values. This results in a modified, but proper penalty matrix. Second, the mixed model representation and reparametrisation described in Fahrmeir et al. (2004) may be used. For example for a penalised spline with second order random walk prior for basis function coefficients, an additional linear effect $\beta_j x_j$ has to be included in the linear predictor while the vector $\boldsymbol{\alpha}_j$ of remaining reparametrised coefficients has a proper ridge-type prior $\boldsymbol{\alpha}_j \sim \mathrm{N}(\mathbf{0}, \delta_j^2 \boldsymbol{I})$. Third, Panagiotelis and Smith (2008) replace the Gaussian smoothness prior (6) through a linearly constrained proper prior, deleting the corresponding unpenalised part from the model. For a P-spline with second order random walk prior, a linear effect has therefore to be added to the predictor. For function selection, they assume a spike and slab prior with a point mass at 0:

$$\boldsymbol{\gamma}_j | \delta_j^2, J_j \sim (1 - J_j)\delta_0(\cdot) + J_j \, \mathrm{N}_{\mathrm{con}}(\mathbf{0}, \delta_j^2 \boldsymbol{K}_j^-), \tag{10}$$

where $\mathrm{N}_{\mathrm{con}}$ denotes a (possibly constrained) Gaussian smoothness prior and $\boldsymbol{K}_j^-$ is a (generalised) inverse of $\boldsymbol{K}_j$. Posterior inference is developed for Gaussian additive models, to circumvent possible pitfalls in connection with this spike and slab prior. As an alternative, which can be extended to non-Gaussian models, we plan to transfer the spike and slab variance (NMIG) approach for variable selection to the problem of function selection. Posterior inference for smooth functions can then be extended to include function selection in conceptually straightforward manner.

3.3 Selection for random effects in mixed models

Variable selection concepts can also be applied to variance component and covariance matrix selection in mixed models. To differentiate more clearly between smooth effects and random effects with non-diagonal covariance matrix and to stress the longitudinal structure of the data, we extend the model equation (3) to

$$\eta_{it} = \boldsymbol{x}_{it}'\boldsymbol{\beta} + \boldsymbol{z}_{it1}'\boldsymbol{\gamma}_1 + \cdots + \boldsymbol{z}_{itr}'\boldsymbol{\gamma}_r + \boldsymbol{u}_{it}'\boldsymbol{\alpha}_i$$

where $\boldsymbol{\alpha}_i \overset{\text{i.i.d.}}{\sim} \mathrm{N}(\mathbf{0}, \boldsymbol{Q})$ correspond to random effects of the covariates in $\boldsymbol{u}$, $i = 1, \ldots, n$ denotes individuals and $t = 1, \ldots, n_i$ observations within individuals. If the components α_{ij}, $j = 1, \ldots, r$, of $\boldsymbol{\alpha}_i$ are assumed to be independent a priori, then $\boldsymbol{Q} = \mathrm{diag}(\delta_1^2, \ldots, \delta_r^2)$, with $\delta_1^2 = \mathrm{Var}(\alpha_{i1}), \ldots, \delta_r^2 = \mathrm{Var}(\alpha_{ir})$. To explore whether a random effect α_{ij} should be included or not, a spike and slab prior for variances can be specified for δ_j^2 in complete analogy to the spike and slab prior of Ishwaran and Rao (2003, 2005) for variable selection. Then, the marginal prior for δ_j^2 is a bimodal mixture of two inverse Gamma densities, replacing, for example $\delta_j^2 \sim \mathrm{IG}(a_j, b_j)$ as a standard conjugate prior for variances of Gaussian random effects.

Selection schemes for diagonal and offdiagonal elements of covariance matrices have been developed by Smith and Kohn (2002) for the multivariate Gaussian distribution, Chen and Dunson (2003) and Frühwirth-Schnatter and Tüchler (2008) for Gaussian linear mixed models, and Cai and Dunson (2006) for generalised linear mixed models. Kinney and Dunson (2007) extend a modified version of Chen and Dunson (2003) to logistic mixed models using an auxiliary latent variable representation. All these approaches aim at parsimonious covariance matrix estimation to avoid unstructured highly parametrised specification through inverse Wishart priors. The common idea is to factorise $\boldsymbol{Q} = \mathrm{Cov}(\boldsymbol{\alpha}_i)$ or its inverse through some type of Cholesky decomposition and to specify appropriate priors for Cholesky factors.

Cai and Dunson (2006) develop a covariance matrix selection method using the Cholesky decomposition $\boldsymbol{Q} = \boldsymbol{D}\boldsymbol{L}\boldsymbol{L}'\boldsymbol{D}$, where $\boldsymbol{D}$ is a diagonal matrix and $\boldsymbol{L}$ is a lower triangular matrix with ones in the diagonal. For selection of elements in $\boldsymbol{D}$ and $\boldsymbol{L}$, they choose a mixture of a point mass at zero and a truncated normal density for the elements d_j^2, $j = 1, \ldots, r$, of $\boldsymbol{D}$, and a mixture of a point mass at zero and a normal density for the elements of $\boldsymbol{L}$. Posterior computation is based on MCMC algorithms in combination with a Laplace approximation of the marginal likelihood of y, integrating out random effects. In view of the potential problems arising from mixtures involving point masses at zero, it seems more natural to employ the spike and slab priors of Ishwaran and Rao (2005) instead. A careful hierarchical specification of such an alternative and an implementation of posterior computation is desirable.

Frühwirth-Schnatter and Tüchler (2008) use the Cholesky decomposition $\boldsymbol{Q} = \boldsymbol{C}\boldsymbol{C}'$ to rewrite the random effects part $\boldsymbol{U}_i\boldsymbol{\alpha}_i$ of the predictor as $\boldsymbol{U}_i\boldsymbol{C}\boldsymbol{b}_i$, where $\boldsymbol{b}_i \sim \mathrm{N}(\boldsymbol{0}, \boldsymbol{I})$. Stacking the elements of $\boldsymbol{C}$ in a vector $\boldsymbol{c}$, the random effects part can be rewritten as $\boldsymbol{W}_i\boldsymbol{c}$, where the design matrix $\boldsymbol{W}_i$ depends on $\boldsymbol{U}_i$ and the latent random effects $\boldsymbol{b}_i$. The basic idea is: If the random effects $\boldsymbol{b}_i$ (and $\boldsymbol{W}_i$) *were known*, then the predictor $\boldsymbol{\eta}_i = \boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{W}_i\boldsymbol{c}$ defines a classical linear Gaussian model, and selection of the matrix is transformed into a variable selection problem for the components of the vector $\boldsymbol{c}$. Given the random effects, Frühwirth-Schnatter and Tüchler (2008) follow the conjugate prior approach with a point mass at zero, assuming a conditional g-prior for $\boldsymbol{c}$. The MCMC scheme includes sampling the latent random effects from their (Gaussian) full conditionals, so that the complete MCMC sampler resembles samplers developed for Bayesian variable selection in Gaussian linear models. This approach may be extended to non-Gaussian mixed models that can be formulated in terms of latent auxiliary Gaussian mixed models, see Tüchler (2008) for logistic mixed models.

4 Applications

To illustrate some of the Bayesian methods reviewed in this paper, we present two applications. The first one focuses on survival times and the regularised estimation of a number of parametric effects. The second application deals with a high-dimensional geoadditive regression model that combines regularisation of a large coefficient vector of parametric effects with nonparametric and spatial modelling. In all cases, the hyperparameters of the Bayesian lasso and Bayesian ridge are set to the weakly informative values $a = b = 0.001$ and the hyperparameters of the Bayesian NMIG are $\nu_1 = 1$, $\nu_0 = 0.000025$, $a = 5$ and $b = 25$. These values are chosen to assign a marginal prior probability of about 0.8 to fall into the interval $[-2, 2]$ to each regression coefficient.

4.1 PBC liver data

Primary biliary cirrhosis (PBC) is an autoimmune disease of the liver which can lead to scarring, fibrosis, cirrhosis, and ultimately liver failure and death of the patient. In this application, we use PBC survival data provided in the R package survival or at the book homepage of Therneau and Grambsch (2000). We refer to this book for a variable description and a detailed frequentist analysis. A total of 424 PBC patients, referred to Mayo Clinic during a ten-year interval (between 1974 and 1984), met eligibility criteria for the randomized placebo controlled trial of the drug D-penicillamine. Discarding observations with missing values leaves $n = 276$ observations with 58.42% right censoring. To make our results comparable to those in Tibshirani (1997), all covariates were standardized to have zero mean and unit variance.

We consider hazard regression models of the form $\lambda_i(t) = \exp(g_0(t) + \boldsymbol{x}_i'\boldsymbol{\beta})$ where $g_0(t)$ refers to the log-baseline hazard rate and $\boldsymbol{x}_i'\boldsymbol{\beta}$ contains effects of covariates. In a full likelihood approach as in Hennerfeind et al. (2006), a cubic penalised spline prior with 20 inner knots and second order random walk prior is assigned to the log-baseline (see Hennerfeind et al. 2006 for the resulting likelihood and numerical details). Alternatively, profiling out the baseline yields partial likelihood approaches as originally considered in the Cox model. As regularisation priors for $\boldsymbol{\beta}$, we used the Bayesian lasso, the Bayesian ridge and the NMIG prior based on 20000 iterations with a burn in of 5000. Thinning the chain by 10 results in an MCMC sample of size 1500. We compare the Bayesian methods with a stepwise-backward procedure for Cox's regression model based on the AIC and frequentist lasso regression provided in the R package penalized (Goeman 2007) where the penalization parameter was determined by n-fold generalized cross validation.

The resulting point estimates, based on both full and partial likelihood, together with the corresponding standard deviations for the regression coefficients, are displayed in the upper plot of Fig. 3. The lower plot of Fig. 3 shows variable selection results for the different regularisation approaches. Since the Bayesian lasso prior does not share the strong variable selection property of the frequentist lasso, hard shrinkage rules are considered to accomplish variable selection. An interval criterion is constructed using the sample standard deviation, so that only regression coefficients with zero outside the one standard deviation interval around the posterior mean are included in the final model. On the other hand, the Bayesian NMIG provides a natural criterion to select covariates if the samples of the indicator variables are utilized (compare end of Sect. 3.1). In our application, we use the intuitive cut value of 0.5 as a selection criterion, i.e. covariates whose corresponding indicator posterior mean exceeds 0.5 are included in the final model. For those coefficients that are not included in the final model, the standard deviations are set to zero.

All methods lead to models that include the five covariates age, alb, bili, stage and copper and eliminate hepmeg, platelet, spiders, trt and trig. The covariate ascites is only chosen by the lasso, but the effect of ascites is very small. Obviously, the NMIG methods shrink small effects to a larger extent than the lasso-based methods, so that most of the remaining covariates (except edrt) are excluded if selection is based on the frequency of the NMIG indicator variables. Figure 4 shows this relative indicator frequencies for the partial and the full likelihood. In Fig. 5, coefficient paths for three selected coefficients from the Bayesian lasso are plotted as a function of the constraint parameter λ together with the pointwise standard deviation. The black solid vertical line marks the estimated constraint parameter $\hat{\lambda}$ and the

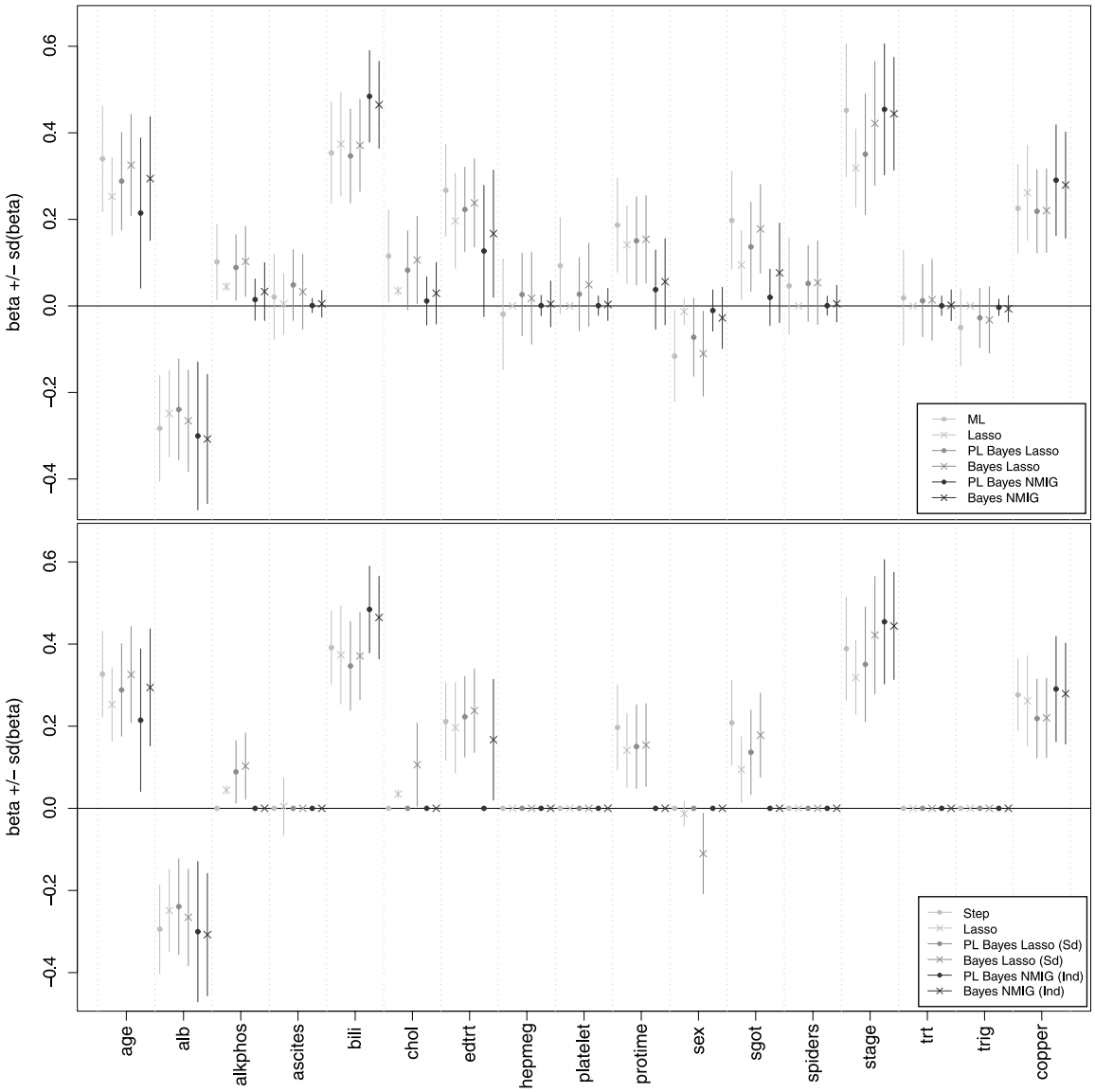


Fig. 3 PBC Liver Data: Estimated coefficients and standard errors without (*upper panel*) and with variable selection (*lower panel*). The methods ML, Step, lasso and PL Bayes are based on the partial likelihood the remaining methods are based on the full likelihood

dashed lines the corresponding 95% empirical credible interval. The movement of the estimator for the covariate chol is of particular interest, since it is not included in all of the final models resulting from the selection procedures. More details of the analysis and simulation results can be found in Konrath et al. (2008).

4.2 Munich rental guide

Most of the German larger cities publish rental guides that provide average rents, obtained from regression models with net rents or net rents per square meter as dependent variable and characteristics of the flat as explanatory variables. In the following, we will analyse data on approximately 3,000 flats in the City of Munich for the 2007 rental guide. The original data (available for research purposes upon request) contain approximately 250 covariates describing characteristics of the flats. While most of the covariates are categorical, the size of the flat (in square meters) and the age of the building (in years) are well-known to have nonlinear impact on the net rent from previous analyses. Moreover, a spatial variable is available that represents the subquarter of Munich where the flat is located. Based on the previous considerations, we will predict net rents per square meter from a high-dimensional geoadditive regression model

$$\text{netrent}_i = \beta_0 + \boldsymbol{x}_i'\boldsymbol{\beta} + f_1(\text{size}_i) + f_2(\text{year}_i) + f_{\text{spat}}(s_i) + \varepsilon_i$$

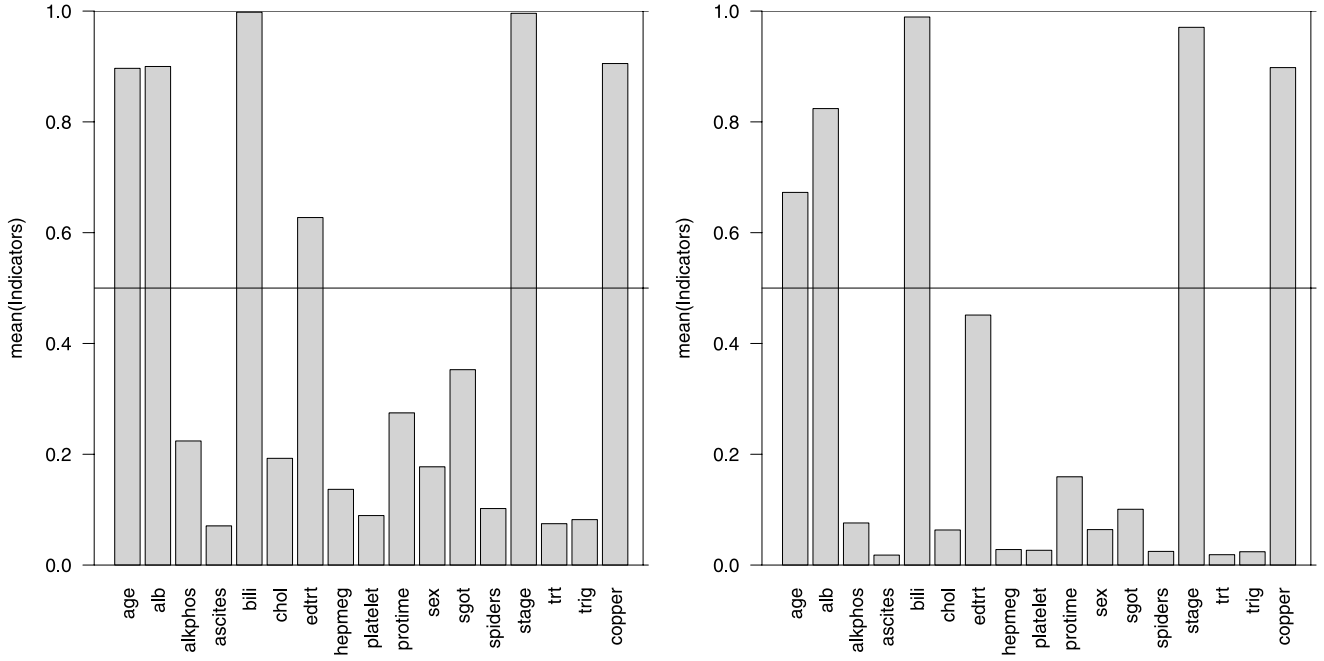


Fig. 4 PBC Liver Data: Relative frequencies of the of the Bayesian NMIG indicator variable value based on the full likelihood (*left*) and the partial likelihood (*right*)

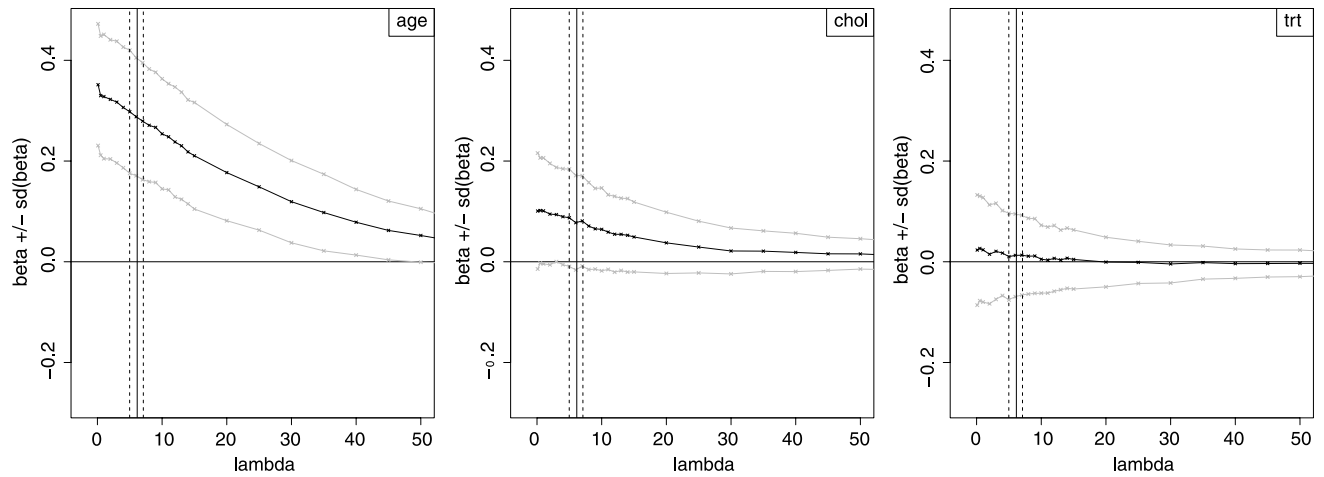


Fig. 5 PBC Liver Data: Selected coefficient estimates based on partial likelihood from the Bayesian lasso (*black lines*) as a function of the constraint parameter together with the corresponding empirical standard deviation (*grey lines*). The *vertical black solid line* marks the estimated value and the *vertical black dashed lines* the corresponding 2.5% and 97.5% empirical quantiles

where β_0 corresponds to the intercept that remains unpenalised, $\boldsymbol{x}_i$ represents a 238-dimensional vector of mostly categorical covariates with effects summarised in the vector $\boldsymbol{\beta}$ that is assigned an appropriate shrinkage prior. To be more specific, we will consider Bayesian ridge, Bayesian lasso and Bayesian NMIG priors. The functions f_1 and f_2 are smooth functions of the continuous covariates size of the flat and year of construction. Both functions are modelled as Bayesian cubic P-splines with twenty inner knots and second order random walk priors. The spatial effect f_{spat} is assigned a Markov random field prior based on the discrete spatial information of the $S = 411$ subquarters of Munich. We use 35000 iterations with a burn in period of 5000 and thin the chain by 30, resulting in an MCMC sample of size 1000.

Figures 6 and 7 visualise the estimated nonlinear and spatial effects. All regularised approaches yield more or less the same results so that only the results for the Bayesian lasso are shown. As expected, high net rents per square meter are observed for smaller flats and there is only very little variation for flats of size larger than about 50 square meters. All models pick up the expected lower net rents for buildings

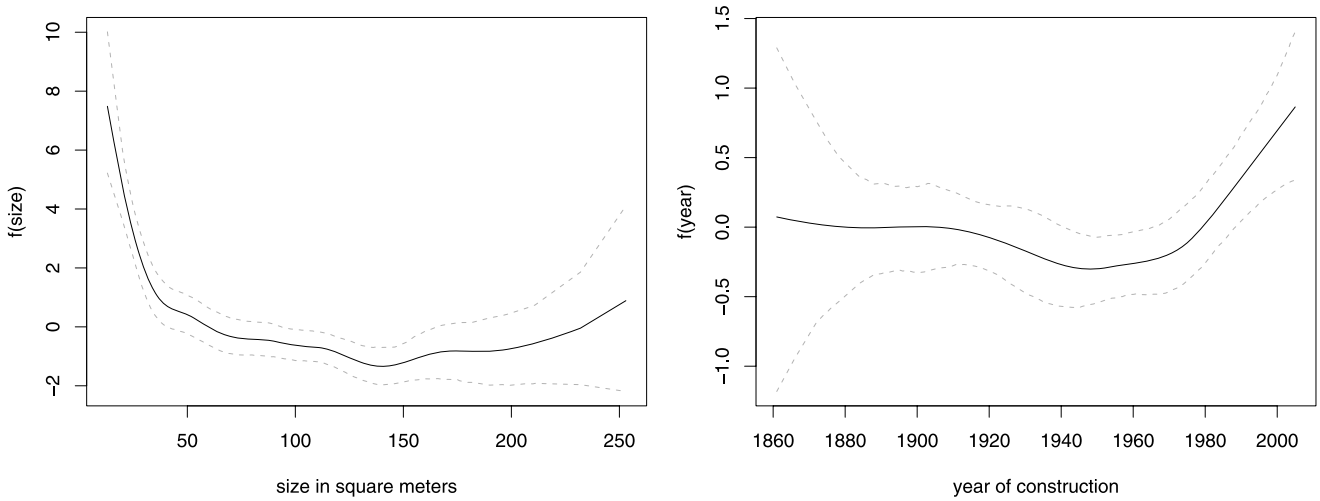


Fig. 6 Munich Rental Guide: Estimated nonparametric effects and 95% pointwise credible bands in the Bayesian lasso model

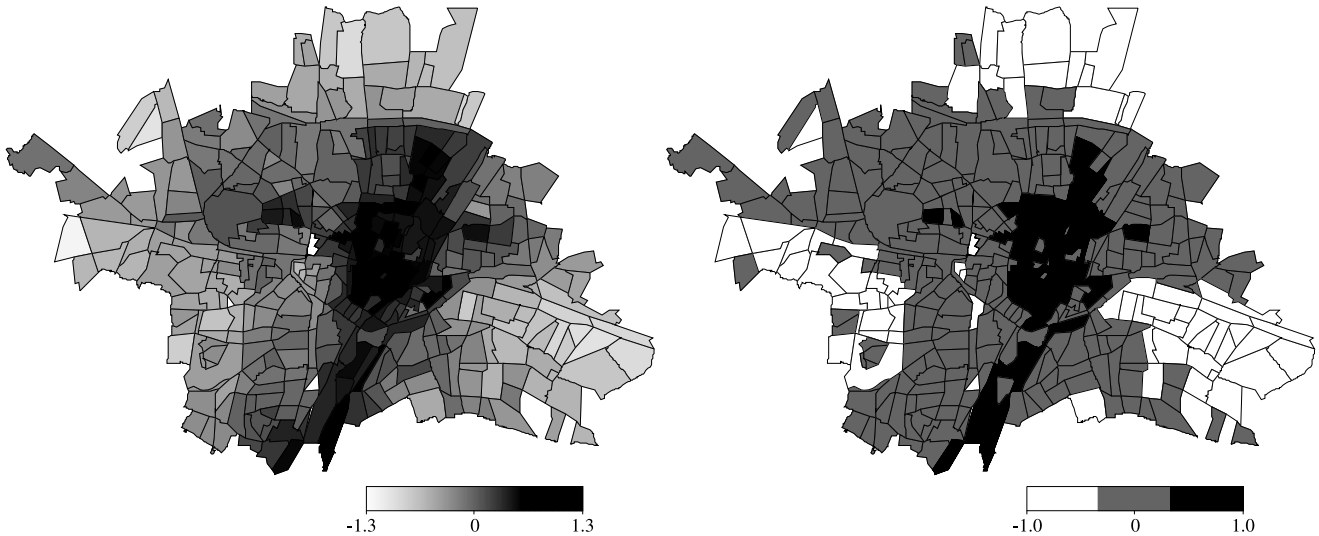


Fig. 7 Munich Rental Guide: Estimated spatial effects and 80% pointwise posterior probabilities in the Bayesian lasso model

constructed during the 1950s to 1970s, while rents start to increase again for newly constructed buildings. The spatial effect identifies increased rents for flats in the center and the southern part of Munich, although several of the covariates already account for the higher quality in the city center and some expensive subquarters in the south. In contrast, significantly negative effects are found for some subquarters at the border of Munich.

Table 1 shows estimated regularised effects for the subset of covariates with associated posterior probabilities smaller than 0.05 in either the Bayesian lasso, the Bayesian ridge, the Bayesian ridge with fixed hyperparameter $\lambda = 0.5$ (leading to weak regularisation of the fixed effects) and the NMIG model together with the corresponding frequencies of the NMIG indicators. While the posterior probabilities are in good agreement between Bayesian lasso and Bayesian ridge, the estimated effects show some differences. In particular, the absolute value of the estimated effects is larger for the lasso in most of the cases where differences of a certain magnitude are observed. This confirms the intuition that the lasso induces less penalisation for larger effects due to the heavier tails of the corresponding prior density. For most of the coefficients the absolute values of the NMIG effects with relative frequencies higher than 0.5 are larger and those with frequencies lower than 0.5 are smaller than the corresponding effects of the Bayesian lasso. It turns out that the dummy for company housing is a variable with very high impact, leading to a reduction of about one Euro for the net rent per square meter. This of course perfectly makes sense since companies will typically rent flats to their employees for a lower charge. The NMIG also selects this covariate in all MCMC iterations, leading to a relative frequency of

Table 1 Munich Rental Guide: Regularised effects with posterior probability lower than 0.05 with either the Bayesian lasso, ridge or NMIG prior or indicators with posterior frequencies exceeding the threshold of 0.5

	lasso		ridge		ridge ($\lambda = 0.5$)		NMIG		
Variable	Mean	Prob.	Mean	Prob.	Mean	Prob.	Mean	Prob.	Ind.
company housing	−1.318	0.000	−0.977	0.000	−1.415	0.000	−1.455	0.000	1.000
fridge	0.436	0.000	0.414	0.000	0.494	0.000	0.589	0.008	0.992
intercom	0.375	0.000	0.362	0.000	0.447	0.000	0.484	0.000	1.000
drying area	−0.429	0.000	−0.427	0.000	−0.433	0.000	−0.497	0.000	1.000
playground	−0.474	0.000	−0.455	0.000	−0.478	0.000	−0.566	0.000	1.000
attic	−0.565	0.000	−0.540	0.000	−0.560	0.000	−0.632	0.000	1.000
no. of occupants	0.119	0.000	0.125	0.006	0.137	0.002	0.057	0.076	0.188
side costs	−0.005	0.000	−0.005	0.000	−0.005	0.000	−0.004	0.000	0.001
floor covering (PVC)	0.545	0.000	0.489	0.000	1.105	0.000	0.954	0.000	1.000
floor covering (felt files)	0.441	0.000	0.389	0.004	1.056	0.000	0.825	0.000	1.000
rent start date before 2002	−0.916	0.000	−0.847	0.000	−0.857	0.000	−1.031	0.000	1.000
tiled bathroom	0.239	0.002	0.250	0.002	0.212	0.016	0.185	0.220	0.542
average residential area	−0.459	0.002	−0.429	0.004	−1.310	0.000	−1.509	0.000	1.000
no waterheating	−1.052	0.004	−0.532	0.004	−0.928	0.124	−2.127	0.000	1.000
no additional heating	−0.375	0.014	−0.400	0.002	−0.422	0.016	−0.379	0.238	0.610
special windows	0.268	0.014	0.272	0.010	0.303	0.006	0.087	0.480	0.235
grassy area	−0.260	0.016	−0.294	0.004	−0.411	0.000	−0.044	0.498	0.127
kitchen size >12 m^2	−0.207	0.022	−0.233	0.006	−0.252	0.002	−0.022	0.540	0.048
drying room	−0.182	0.022	−0.202	0.006	−0.211	0.010	−0.083	0.296	0.279
bath width	0.139	0.024	0.160	0.018	0.117	0.086	0.033	0.334	0.072
no. roof terraces	0.506	0.026	0.406	0.024	0.870	0.006	1.132	0.014	0.985
floor covering (simple wood)	0.337	0.026	0.312	0.010	0.980	0.000	0.793	0.000	1.000
modernised windows	−0.161	0.028	−0.189	0.024	−0.223	0.006	−0.011	0.628	0.008
modernised floor	0.183	0.030	0.197	0.024	0.171	0.060	0.036	0.456	0.094
bathtub	0.260	0.032	0.297	0.018	0.287	0.046	0.014	0.736	0.028
room 3 facing garden	0.219	0.040	0.247	0.054	0.363	0.070	0.082	0.418	0.229
built-in kitchen	0.215	0.044	0.236	0.030	0.231	0.038	0.044	0.542	0.109
2nd bathroom	0.327	0.048	0.327	0.030	0.498	0.012	0.514	0.140	0.769
shower	0.216	0.054	0.254	0.006	0.240	0.036	0.011	0.672	0.014
no balcony	−0.207	0.056	−0.242	0.036	−0.202	0.130	−0.014	0.632	0.028
satellite dish	−0.253	0.060	−0.273	0.022	−0.400	0.000	−0.052	0.554	0.120
room 4 facing street	−0.290	0.060	−0.300	0.032	−0.675	0.024	−0.109	0.550	0.227
cooking hobs	0.231	0.064	0.262	0.026	0.309	0.020	0.051	0.590	0.132
room 1 facing south	0.173	0.064	0.219	0.016	0.384	0.000	0.012	0.622	0.009
stairwell renovation	−0.175	0.066	−0.205	0.020	−0.197	0.026	−0.014	0.636	0.022
penthouse appartment	0.279	0.076	0.297	0.024	0.337	0.058	0.011	0.882	0.027
electric/gas stove	−0.220	0.078	−0.240	0.054	−0.346	0.038	−0.237	0.188	0.678
floor covering (linoleum)	−0.248	0.088	−0.313	0.010	0.285	0.134	0.001	0.986	0.014
2nd toilet	0.187	0.100	0.199	0.048	0.221	0.074	0.053	0.582	0.140
video interphone system	0.339	0.154	0.296	0.132	0.905	0.004	0.798	0.230	0.717
ground floor flat	−0.342	0.268	−0.262	0.246	−1.912	0.000	−2.090	0.064	0.937
gas furnace	−0.195	0.274	−0.190	0.294	−0.165	0.758	−0.353	0.304	0.611
floor covering (felt)	0.196	0.314	0.194	0.264	0.953	0.000	0.852	0.108	0.894
good address	−0.146	0.394	−0.125	0.382	−1.010	0.000	−1.155	0.000	1.000
basic address	−0.218	0.400	−0.212	0.324	−1.913	0.000	−2.585	0.008	0.991

one for the corresponding indicator. Several other covariates included in the list in Table 1 correspond to specific equipment of the flat such as different types of floor material or the availability of an intercom system. Remarkable are covariates like basic address or ground floor flat. The Bayesian lasso and Bayesian ridge estimate a small effect with large posterior probabilities but the NMIG effect is in contrast larger and much closer to the effect estimated by Bayesian ridge with fixed λ. The corresponding indicators also show high inclusion frequencies.

5 Conclusions

Our focus has been on Bayesian regularisation through conditionally Gaussian priors in high-dimensional structured additive regression models, providing a unified framework for modelling and statistical computing. We have deliberately omitted approaches based on Dirichlet process (DP) or related priors as another popular direction in nonparametric regression, see e.g. Müller and Quintana (2004) for a survey. Our main reason for this has been that these approaches are less suited for regularizing high-dimensional regression models. Rather, they are useful for relaxing restrictive assumptions on distributions of error terms, random effects or frailties by allowing more heterogeneity. DP based priors may be combined, however, with modelling concepts using Gaussian smoothness priors or adaptive selection of basis functions, see e.g. Li et al. (2009) and Bigelow and Dunson (2007) for recent work. In our own research, we currently develop models for longitudinal data incorporating Bayesian P-splines for nonlinear trends and DP (mixture) priors for random effects.

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